

Project Design Phase
Proposed Solution Template

Date	29 JUNE 2026
Team ID	SWTID-2026-7871
Project Name	India's Agricultural Crop Production Analysis
Maximum Marks	2 Marks

Proposed Solution :

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Agricultural Production data in India is vast and spread across multiple states, crops, seasons and years. It is difficult for farmers, researchers, and policymakers to analyze long-term trends(1997-2021) and derive meaningful insights quickly. There is a need for an interactive visualization solution that can provide a clear overview of crop production performance to support data- driven decision making.
2.	Idea / Solution description	The proposed solution uses Tableau to analyze India's agricultural crop production data from 1997 to 2021. Interactive dashboards and stories are developed to visualize production trends, state-wise crop performance, seasonal variations, yield patterns, and comparative analysis. The solution enables users to explore data through filters, KPIs, charts, maps, and storytelling dashboards.
3.	Novelty / Uniqueness	• Analysis of 25 years of agricultural production data. • Interactive dashboards with dynamic filters for State, Crop, Year, and Season. • Combination of maps, KPIs, trend analysis, and comparative visualizations. • Tableau Story feature used to provide data-driven narratives. • Publicly accessible dashboards through Tableau Public for wider reach and transparency.
4.	Social Impact / Customer Satisfaction	• Helps farmers understand crop production trends and make informed decisions. • Assists policymakers in agricultural planning and resource allocation. • Supports researchers and students in agricultural data analysis. • Improves accessibility and transparency of agricultural information. • Provides an easy-to-use interface with interactive visualizations for enhanced user experience.
5.	Business Model (Revenue Model)	This project is developed for academic purposes and published using Tableau Public. In a real-world scenario, the solution could follow a subscription-based or freemium model, where advanced analytics,

		predictive insights, and customized reports are offered to government agencies, research institutions, agri-tech companies, and agricultural organizations
6.	Scalability of the Solution	• Tableau can efficiently handle large datasets and complex visualizations. • Additional years, crops, states, and agricultural datasets can be integrated. • Future enhancements may include predictive analytics and machine learning models. • Weather, soil quality, and market price datasets can be incorporated. • The solution can scale from regional analysis to national and global agricultural studies.